

Movender: Design of an Interactive Mobile Vending Robot

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https://github.com/Shiyizhuanshi/HCR_Movender.git

Abstract—Movender is an AI powered robotic merchant whose purpose is to promote products such as snacks and beverages. It can be deployed both indoors and outdoors to actively look for potential customers, and then use human-friendly interaction to recognize people’s intention in order to provide the correct service. The robot is equipped with a navigation system, a speech recognition module, a computer vision module and a system for user-robot interaction. This robot is the main tool for our research project into whether increasing active interaction between the robot and the human will increase the number of sales and customer satisfaction. In this report, we have described our main theory, the design details of the overall system and the approaches we’ll take to evaluate our experiment.

I. INTRODUCTION AND MOTIVATION

Traditional vending machines are “passive vendors”, they cannot move or actively influence customer decisions. There are several studies on increasing the efficiency of vending machines:

- Strategic placement: The strategic placement of vending machines in high-traffic areas capitalizes on consumers’ tendencies to make spontaneous, unplanned purchases.[1]
- Consumer psychology: The consumers mindset could be manipulated to nudge them for better food choices. [2]

However, these studies are all based on the reality that the traditional vending machines are passive. Even with all these efforts being investigated into traditional vending machines, they still have problems being a passive vendor:

- Even when the vending machines are at obvious places, a customer might fail to spot the machine. When customers want to buy something from the machine without a machine in sight, they might fail to remember where the last one they saw were, and it might be too far away.
- Even when the vending machine is passively trying to influence the customer decision, only a small fraction buy the influence. It cannot actively convince someone to buy their product.
- Restocking vending machines require someone to visit all of the machines. At a large scale, it is both manpower consuming and time consuming.

From the related work section below, we can see that there is an increasing global awareness on interaction-active robots. In this project we propose an active-vendor, a traveling and intelligent vending robot, Movender. Who is able to spot the crowd, navigate to the crowd, pitch its products, and vary its

conversation based on the customer’s facial expression. We believe this effectively addresses the problem of traditional passive vending machines outlined above.

II. RELATED WORK

While there is limited research on vending robots, relevant work exists in the broader domain of service robots. Several studies explore the integration of mobility, human interaction, and task-specific functionalities in service robotics.

A. Service Robots

Systematic literature reviews [3] highlight the increasing adoption of service robots across various sectors, classifying them into professional and domestic domains based on interaction capabilities and environments. These studies emphasize key features such as autonomous navigation, social interaction, and human-centered design.

1) *CMU Snackbot*: The *Snackbot* [4] was designed for snack delivery in office environments, focusing on user-centric design and social interaction. It incorporated a navigation system and dialogue-based interaction. However, its scope was limited to indoor environments, restricting its exposure to diverse human behaviors and real-world vending scenarios.

2) *Dobby*: *Dobby* [5], a conversational service robot powered by GPT-4, was tested in tour-guide scenarios, responding to human commands and interacting with the physical world. Experiments showed that users preferred the conversational robot over a non-conversational counterpart. However, improvements could be made by reducing response delays and incorporating nonverbal cues such as facial expressions. This study is relevant to our work as it demonstrates how integrating large language models (LLMs) enhances human-robot interaction.

3) *SPENCER*: The *SPENCER* robot [6], deployed at Schiphol Airport, assisted passengers by guiding them from arrival gates to key locations. It showcased robust navigation in dynamic public spaces, leveraging social awareness to handle busy areas and understand group interactions. This capability aligns with our robot’s need to operate efficiently in crowded outdoor environments.

While service robots commonly employ computer vision for navigation and social behavior analysis, its application in sentiment analysis remains underexplored. Enhancing robots with affective computing could significantly improve human-robot interaction efficiency.

B. Affective Computing

Affective computing enables robots to recognize and respond to human emotions by analyzing multimodal cues such as facial expressions, voice tone, and gestures [7]. While traditional sentiment analysis focused on text, multimodal approaches incorporating visual and audio data have significantly improved accuracy [8].

Tensor Fusion Networks (TFN) enhance robotic affective computing by modeling both *inter-modality interactions* (how speech, visuals, and acoustics combine to convey sentiment) and *intra-modality variations* (how emotions evolve within each modality) [8]. This allows robots to dynamically interpret user reactions, improving engagement and adaptability [7].

Real-time sentiment analysis enables social robots to refine their responses based on user feedback, making interactions more natural [7]. Studies confirm that fusing visual, audio, and text inputs enhances emotion detection, improving robot behavior [8].

Facial expressions play a key role in emotion recognition. *OpenFace*, an open-source toolkit, enables real-time analysis of facial action units (AUs), head pose, and eye gaze without requiring specialized hardware. Integrating such tools allows robots to adapt interactions based on user emotions, further enhancing engagement [9].

While existing service robots excel in specific domains, none integrate *mobility, LLMs, affective interaction, and vending functionalities* into a single system. Our work aims to bridge this gap by combining *real-time sentiment analysis, advanced human-robot interaction, and autonomous vending* into a unified mobile platform. By incorporating insights from prior research, we seek to develop a vending robot that not only dispenses products but also engages with users effectively, enhancing the overall experience.

III. HYPOTHESIS

The study aims to investigate the effect of interactivity on the percentage of successful sales. This investigation allows us to train an optimal interactive model for mobile vending robots, maximizing its functionality.

We propose the main hypothesis: ***The more interactivity between the robot and the customer, the higher percentage of sales will be successful.***

We can define interactivity in a spectrum, from low to high:

- Low interactivity: There is only minimal interaction between the customer of the robot, like a traditional vending machine. The machine response passively to the user's input.
- Some interactivity: There exists some form of voice interaction, but the machine is only responding with a preset fixed language base, with no variations regardless of customer behavior.
- High interactivity: The robot actively approaches customers, and change its pitching behavior with respect to the user input. For example, when the robot detects the user has low interest in the product, it uses another persuasion technique.

IV. SYSTEMS DESIGN

The robotic platform selected for the project is the Pioneer 3-AT. This platform is comprised of various hardware sensors to support navigation, interaction with the environment, and human-robot communication. A wheel encoder is integrated into robot's built-in system to maintain essential odometry data that supports its positioning and mobility. An additional LIDAR sensor can handle a high-resolution scan of the environment and assist with obstacle detection, while an IMU sensor can sense directions, angular velocity, and linear acceleration. Stereo depth cameras is also employed to serve the purpose of enabling the robot to detect human faces and emotion changes. A microphone allows the robot to command from a human's voice, while a loudspeaker could facilitate voice interaction, better communication and enhanced task interoperability with humans.

A. Navigation

1) *Challenges*: Autonomous navigation in human environments is a challenging problem, as mobile robots must maneuver safely through crowds of unpredictable pedestrians [10]. The robot should avoid collisions while efficiently reaching its destination, requiring real-time perception and planning under dynamic conditions.

The Movender robot is built on a Pioneer 3-AT (P3AT) mobile base and is designed as a merchant platform that can operate in busy public spaces. Its navigation objectives include safe passage through crowded areas to deliver goods or services, while respecting personal space and ensuring human safety.

2) *System Architecture*: The Movender's navigation framework follows a typical layered architecture, consisting of mapping, path planning, and motion control modules. This design is common in mobile robotics, where simultaneous localization and mapping (SLAM), global/local path planning, and low-level control work together for autonomous navigation [11].

In our system, a *GridManager* module builds and maintains a local occupancy grid map of the environment, continuously updated with sensor data to reflect nearby obstacles. On top of this map, a *PathPlanner* module computes collision-free routes using the A* search algorithm, taking into account obstacle clearance and the robot's kinematic constraints. Finally, a *RobotController* module handles motion execution, using a proportional–integral–derivative (PID) controller to follow the planned path and adjust the robot's velocities. These components communicate through the Robot Operating System (ROS) middleware: the robot's sensors (e.g., a 2D LiDAR scanner) feed data to the GridManager, the PathPlanner plans a path when a new goal is received, and the RobotController continuously drives the wheel motors based on the latest plan. This modular architecture allows the Movender robot to perceive its surroundings, plan safe paths, and move appropriately in a closed loop. It also facilitates debugging and enhancements, allowing each subsystem (mapping, planning, control) can be developed and tested independently before integration.

3) *Grid-Based Occupancy Mapping*: The Movender relies on a grid-based map to represent obstacles in its vicinity. The GridManager maintains a 2D occupancy grid centered on the robot, discretizing the environment into grid cells (e.g., 0.2 m resolution over a 10×10 m area). Each cell holds a boolean state (occupied or free) indicating whether an obstacle was sensed there. This occupancy grid is updated in real-time from incoming LiDAR readings: each laser scan point is transformed into the robot's coordinate frame and mapped to a grid cell, which is then marked occupied. A classic occupancy grid approach assumes a static environment [12]; however, human crowds are highly dynamic, so our GridManager implements mechanisms for dynamic occupancy handling. First, it tags obstacle cells with timestamps and periodically decays old obstacles: any cell occupied longer than a persistence threshold (e.g., 0.2 s) without new detections is cleared. Second, the GridManager enforces a clear zone around the robot, keeping cells immediately surrounding the robot's position free to avoid false positives from the robot's own body or sensor noise. When the robot moves (tracked by odometry), the GridManager shifts or rotates the grid to maintain a consistent local map.

4) *Path Planning*: Once the environment is represented in the occupancy grid, the PathPlanner computes a safe path to the target location. Path planning is a critical capability for autonomous robots, defined as constructing a safe, collision-free route from a start point to a goal [11]. The Movender uses an A* search algorithm operating on the grid map to find an optimal path, enhanced by several considerations for navigation in human environments:

- **Obstacle Inflation**: The occupancy grid is converted into a cost map where obstacles are expanded to account for the robot's radius and a safety buffer [13], [14].
- **Heuristic and Cost Function**: A heuristic combines Euclidean distance to the goal with an angular cost term, favoring smoother, straighter paths over zig-zagging, leading to more natural trajectories [10].
- **A* Search Process**: Starting from the robot's current cell, the algorithm iteratively expands nodes based on lowest cost, returning a waypoint-based path.

The combination of obstacle inflation, heuristic guidance, and path smoothing provides a path planning strategy that is both safe and efficient. The system dynamically replans if paths become obstructed or goals change.

5) *Motion Control & Navigation Execution*: The Robot-Controller module converts the planned path into smooth velocity commands (linear and angular velocities) that drive the robot along the path. Key aspects include:

- **Waypoint Tracking with PID Control**: PID controls angular velocity for smooth turning towards waypoints.
- **Dynamic Map & Path Update**: Updates to the environment map and path waypoints according to robot's movements.
- **Emergency Stop**: If obstacles appear unexpectedly close, the system triggers immediate high-frequency replanning and stops the robot to avoid collision.

6) *Visualization and System Monitoring*: The Movender leverages ROS visualization tools (RViz) to provide real-time insights, producing visual markers including:

- Robot Pose (central arrow)
- Occupancy Grid Display (red grid)
- LiDAR Scan Points (red points)
- Planned Path (green line & blue waypoints)
- Safety Zones (cyan rectangular outline)
- Emergency Area (red circle)

An example visualization is shown in Figure 1.

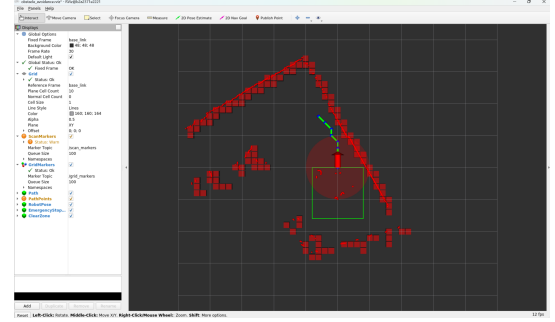


Fig. 1. Movender visualization in RViz.

Debug Pose Publisher for Simulation: A utility simulating robot pose and sensor updates, facilitating offline testing and algorithm verification without hardware.

7) *Conclusion*: The Movender's navigation system demonstrates a practical approach to autonomous navigation in human environments by integrating occupancy grids, A* search, and PID control with crowd-aware features. It ensures safety and reliability, laying a foundation for future advanced social navigation capabilities.

B. Computer vision

The computer vision (CV) module is a critical component of our robot, designed to enhance human-robot interaction by performing real-time person detection (YOLOv8), behavioral analysis, and engagement evaluation. The system integrates an OAK-D camera with holistic analysis based on Mediapipe, enabling robust interpretation of human actions such as pose, gesture, and emotion recognition.

1) Hardware Setup:

- **OAK-D Camera**: Positioned at the top front of the vending robot to maximize the field of view (FOV) and minimize occlusions. This camera provides both RGB images and depth maps, enabling robust person detection and tracking even in dynamic, crowded environments. It has an adjustable detection range of up to 100m, making it perfect for indoor and outdoor detection tasks.
- **Raspberry Pi**: A Raspberry Pi is used to handle real-time data processing. It efficiently manages the data throughput from the OAK-D camera and runs the integrated computer vision algorithms.

2) *Software Architecture*: The computer vision system follows a modular pipeline, ensuring flexibility and scalability. The pipeline consists of the following key stages:

- **Data Acquisition:** Captures real-time RGB and depth data from the OAK-D camera. The data acquisition process uses a receive and queue function that obtains detection frames with fixed data size. The DepthAI pipeline provides fast execution through multiple parallel thread processing, it can also be customized to balance between speed and computation power.
- **Object Detection:** YOLOv8 is employed for robust person detection. It provides bounding boxes and spatial coordinates (with depth information) for detected individuals, allowing the system to focus subsequent analysis on relevant regions.
- **Holistic Analysis:** For each detected person within a predefined close-range threshold (e.g., 1.5 meters), the system performs detailed analysis using a customized Mediapipe holistic model. This module extracts face, hand, and pose landmarks and computes key metrics for further behavioral evaluation.
- **Behavioral Scoring:** The system computes multiple scores based on:
 - **Emotion Detection:** Analyzes facial landmarks to determine emotion (Surprised, Happy, Neutral, Disgusted) and assigns corresponding scores. The facial landmarks are a series of coordinates that marks the spatial locations of the facial features, such as the nose, eye brow, mouth and lip. We can then use distance metrics to calculate the relative movements of facial landmarks. These can then be used to estimate different emotions based on eye brow positions, mouth openness, eye openness, and so on.
 - **Gesture Recognition:** Evaluates hand landmarks to distinguish gestures (Thumb up, Indexing, Hi, Rejecting) and maps them to specific scores. Similar to facial landmarks, the hand landmarks are normalized and the relative distance between the finger tips and the wrist are calculated. These were used to predict different gestures using the principal finger distance.
 - **Pose Estimation:** Assesses body posture via key-point ratios to determine openness or closedness, contributing to the overall engagement evaluation. A mid-body point is used to determine if the hand is raised above the center height of a human figure. This is used to filter out hand gestures that due to free hanging. The coordinates of the head is used to determine whether a customer's head is pointing towards or away from the robot.
 - **Additional Metrics:** Includes gaze direction (estimated using the head pose coordinates calculated previously), head orientation, and hand movement analysis.

These scores are combined with a distance-based score (derived from depth data) to produce an aggregated CV score for each detected person.

- **Output Integration:** The system updates a shared state dictionary with key interaction parameters such as engagement status, the coordinates of the highest score person (goal position), and the overall CV score. This

data can be further integrated with the robot's navigation and interaction modules.

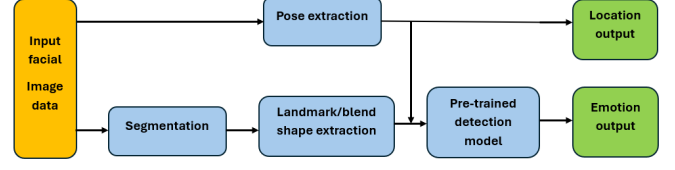


Fig. 2. Overview of the computer vision system.

3) Purchase Intent Estimation:

• Overview

This module estimates a user's purchase intent by integrating multiple behavioral factors, including facial expressions, body posture, hand gestures, gaze tracking, head orientation, and hand movement. Each detected person is assigned a priority score, helping the vending robot prioritize interactions.

• Methodology

Our approach leverages DepthAI's YOLO spatial detection for identifying human subjects and Mediapipe for extracting meaningful behavioral features. Each detected person is assigned a Purchase Intent Score S_{total} , computed as follows:

$$S_{total} = \alpha S_{distance} + \beta S_{emotion} + \gamma S_{pose} + \delta S_{gesture} + \epsilon S_{gaze} + \zeta S_{head} + \eta S_{hand} \quad (1)$$

where:

- $\alpha, \beta, \gamma, \delta, \epsilon, \zeta, \eta$ are weight factors tuned experimentally.
- $S_{distance}, S_{emotion}, S_{pose}, S_{gesture}, S_{gaze}, S_{head}, S_{hand}$ represent different behavioral factors contributing to purchase intent.

• Scoring Metrics

1. Distance Score ($S_{distance}$)

Buyers closer to the vending robot are prioritized:

$$S_{distance} = \frac{2}{d'}, \quad d' = \sqrt{X^2 + Z^2} \quad (2)$$

where (X, Z) are the real-world coordinates of the detected person.

2. Emotion Score ($S_{emotion}$)

Based on facial expression classification:

$$S_{emotion} = \begin{cases} +2, & \text{if Surprised} \\ +1, & \text{if Happy} \\ 0, & \text{if Neutral} \\ -1, & \text{if Disgusted or Negative Expression} \end{cases} \quad (3)$$

3. Pose Score (S_{pose})

Body posture openness and engagement:

$$S_{pose} = \begin{cases} +2, & \text{if pose ratio} > 1.5 \quad (\text{open posture}) \\ 0, & 0.8 \leq \text{pose ratio} \leq 1.5 \quad (\text{neutral posture}) \\ -2, & \text{if pose ratio} < 0.8 \quad (\text{closed posture}) \end{cases} \quad (4)$$

4. Gesture Score ($S_{gesture}$)

Hand gestures indicating purchase intent:

$$S_{gesture} = \begin{cases} +1.5, & \text{if Indexing (actively selecting)} \\ +1, & \text{if Hi (greeting)} \\ +0.5, & \text{if Thumbs up (positive but passive)} \\ -1, & \text{if Rejecting (dismissive gestures)} \end{cases} \quad (5)$$

5. Gaze Tracking Score (S_{gaze})

Indicates whether the customer is looking at the vending machine:

$$S_{gaze} = \begin{cases} +1, & \text{if looking at the vending machine} \\ 0, & \text{if looking away} \end{cases} \quad (6)$$

6. Head Orientation Score (S_{head})

Measures engagement based on head direction:

$$S_{head} = \begin{cases} +1, & \text{if head facing towards the vending machine} \\ 0, & \text{if neutral} \\ -1, & \text{if head turned away} \end{cases} \quad (7)$$

7. Hand Movement Score (S_{hand})

Evaluates whether the user's hand is moving towards the vending machine:

$$S_{hand} = \begin{cases} +1, & \text{if hand approaching vending machine} \\ 0, & \text{if hand stationary or neutral} \\ -1, & \text{if hand moving away} \end{cases} \quad (8)$$

4) *Customer tracking*: The Movender robot needs to travel around the room or street and detect potential customers to promote the product. This can be done easily using the priority score (or the buying score) calculated using the method outlined in the previous section. The DepthAI camera has a built in stereo camera pair [15], [16] which automatically estimates the distance from an object to the camera. In our experiment, we have set the label to only detect people and for all other objects the robot will regard them as obstacles. This will be covered in the navigation section. The stereo camera is integrated with a yolo8 spatial detection model that simultaneously creates a bounding box around each detected person and computes their depth from the camera. For our robot, we have to types of customer tracking strategy:

1) Single person detection

This strategy is used when there're not much people within the FoV, for example in a room. In this case, the person with the highest priority will be approached and interacted.

2) Crowd detection

This strategy is most suitable for detecting people outdoors, for example, in the streets where people are more likely to be gathering in groups. After estimating the spatial coordinates of every people within view, a K-means clustering algorithm is used to divide people into groups. The priority score of each group will be the sum of the priority scores of every individual within the group. Similar to before, the robot will approach the group with the highest priority score. After this, the single person score detection strategy will be used to obtain the individual with the highest priority score within that group and that person will be the first to be interacted by the robot.

Interaction strategy In our final implementation, the emotion and gesture scores are only calculated when the customer is close to the robot. This is to provide more accurate detection and, more importantly, to provide better interaction with the customer who're close. For each person detected:

- The distance score and pose score are first used to determine whether the robot should approaches a particular person.
- As the robot moves towards the target person, the distance between them shortens until a distance threshold is reaches. This is when the robot will all the intention detection including emotion and gesture to predict the customer's willingness of buyin and the degree of satisfaction of the customr with the product or service.

C. Mechanical Design

The main objective of the mechanical design Movendor is to make it visually attractive to customers while maximizing user accessibility. We took inspiration from the Villagers in the game Minecraft,[17] they are in-game friendly creatures that would trade "emeralds" (a kind of currency) for goods. Almost everyone at Imperial has either seen or played Minecraft before; a walking villager would be enough to attract customers. It is also made of rectangular blocks, which we can easily make and assemble.

It is designed to have its arms as the goods dispenser, its nose as the card reader, its mouth holding the speaker. A camera mount is mounted on the head with a microphone attached. The inner structure is assembled using wood, and covered with an acrylic shell. The head contains the raspberry pi and other components. An ipad is mounted on the head to use as display. The whole physical design is modeled using fusion360. Woodworking is done in Imperial Hackspace, laser cutting and 3d printing is done in EEE building level 1 lab.

D. Hardware Development: Vending Mechanism

The Vending mechanism is responsible for delivering purchased goods to the customer in an efficient and reliable manner. This section outlines the hardware design, the selection of actuation methods, and the implementation details of the delivery system.

The Vending mechanism is designed with storage compartments on both sides of the robot. The items are stored in



Fig. 3. How the robot looks

these compartments and pushed to a front delivery space upon purchase. The dispensing of goods is facilitated by a **linear actuator**, which provides precise and controlled movement to ensure smooth item transfer. The actuator extends to push an item from the storage area to the front retrieval area, where the customer can easily access it.

This mechanism was specifically chosen due to **safety considerations**. Using a linear actuator rather than robotic arms prevents direct human interaction with the internal storage and dispensing system, reducing the chances of accidental injury or mishandling.

The key components of the selling mechanism include:

- **Linear Actuator:** Used to push items from the storage space to the front delivery area. The actuator provides precise control over movement, ensuring consistent and reliable item delivery.
- **Storage Compartments:** Located on both sides of the robot, these compartments securely hold the products until a purchase is made.
- **Return Mechanism:** The actuator retracts after each delivery to reset the system for the next transaction.

The linear actuator is composed of several key components: a servo motor, a gear, a motor holder, and a bar that applies force to the item. We chose a servo motor over a stepper motor due to its lower power consumption, which is crucial for the efficiency of this application. A speed servo motor was specifically selected because it allows the actuator to cover a greater displacement range, unlike an angle servo motor, which is limited to smaller movements. The remaining components of the actuator, such as the motor holder and the bar, are custom-designed and fabricated using 3D printing. This manufacturing method enables precise and tailored parts, ensuring a perfect fit within the overall system.

The detailed operation process is given below:

- 1) A customer selects and purchases an item through the robot's interface.
- 2) The system triggers the actuator corresponding to the selected item.
- 3) The actuator extends, pushing the item from the side storage to the front retrieval area.

- 4) The customer collects the item from the designated front space.
- 5) The actuator retracts to its default position, ready for the next transaction.
- 6) For sample goods, a similar mechanism is used, either triggered manually by the customer or automatically at predefined intervals.

1) Considerations and Challenges:

- **Force and Stroke Length:** The actuator must be carefully selected to provide adequate force for pushing items with different orientation to prevent damaging the items.
- **Item Retention Mechanism:** A proper size of hole should be designed to prevent accidental movement of items before they are pushed by the actuator or unexpected item stuck.

To overcome these challenges, we invested considerable time and effort in a process of trial and error. This involved several mechanical redesigns aimed at enhancing the overall functionality of the system, including modifications to the item storage method and adjustments to the shape and design of the actuator. We also spent time fine-tuning the speed and timing of the servo motor's operation, adjusting these parameters to ensure that each item is delivered smoothly and reliably. Through this iterative process, we were able to optimizing the performance of the actuator for seamless item delivery.

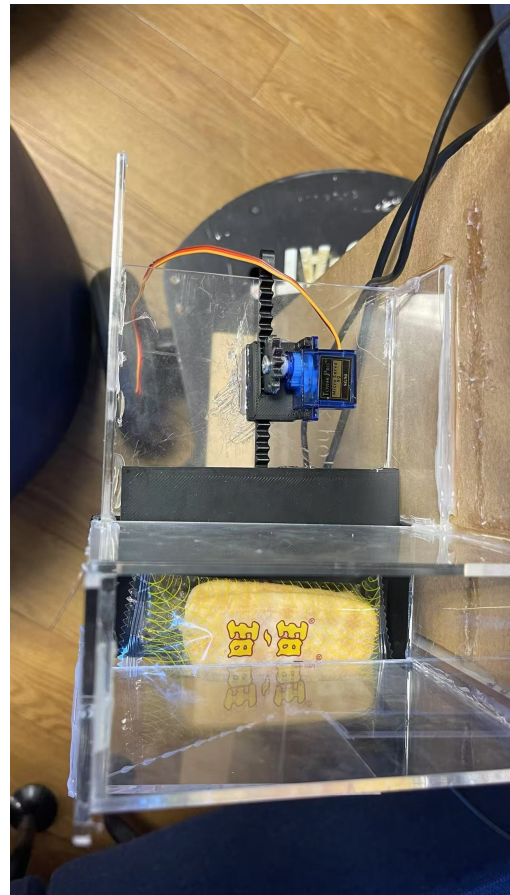


Fig. 4. actuator (approximately to scale)

The final design is given in 4, the cookie will be stacked

upward in the storage space, and there is a plate that provides enough contact to push the cookies, once a cookie is pushed out, the rest of the cookies would slide down due to gravity.

E. UX/Interaction

The interactive user interface (UI) was created to enhance users' interaction with Movendor. The UI design was largely based on HTML, CSS, and JavaScript, with an emphasis on making navigation easy, providing visual clarity, and ensuring responsiveness. Some of the main features like product catalogs, cart options, and real-time updates are given prominence in the main interface so that users can have a smooth shopping experience.

JavaScript was used heavily to control shop interactions dynamically. Product rendering is controlled by 'generateShop', cart updating by 'updateCart', and summarizing by 'generateCartSummary' respectively, with seamless, real-time updating between pages. Real-time communication was also boosted through the addition of Socket.IO, allowing real-time updating of cart items and payment status without page reloading.

The CSS design promoted an accessible, clean, and good-looking appearance. The interface utilizes principles of responsive design, implementing flexbox and media queries to ensure usability across different device sizes. The shopping cart sidebar, product frames, and navigation bar were crafted with strict regard for usability and consistency.

The flow of operations was arranged in a sequential manner over several HTML pages: the main product page ('index.html'), the checkout page ('checkout.html'), the payment processing page ('payment.html'), and the transaction success page ('success.html'). The ordered flow is intended to promote a user-oriented transaction process, eventually reaching the end of an automated vending experience with the successful confirmation of payment.

This part draws inspirations from design principles of human-computer interaction, real-time data communication, and responsive web design. Based on these design approaches, this system enhances operating efficiency and greatly enhances user satisfaction through the delivery of responsive and intuitive interactions.

F. Voice/NLP Design

The main task of voice in human-centered robotics project is to create the voice interaction system, allowing free and interactive communication between the vending robot and customers. The overall goal is to achieve a seamless user experience by precisely recognizing customer speech inputs, determining their meaning, and producing proper responses from the robot.

For automatic speech recognition, the Vosk speech recognition model was employed due to its compact size and capability of operating on resource-constrained hardware, Raspberry Pi 5. Real-time processing of audio from the onboard microphone (ReSpeaker USB 4MIC Array) was enabled by this library, translating customer speech directly into text. Keyword-based detection was implemented to recognize frequently used

expressions, including "who are you," "what do you sell," "no," and "thank you," allowing instant responses.

To facilitate efficient and clear voice output, the Piper text-to-speech engine was integrated. This integration allowed robot responses to be delivered through pre-rendered audio files, significantly reducing latency compared to run-time speech synthesis. Simultaneous audio tasks—particularly transitions between passive advertising and active interaction modes—were managed via Python's subprocess module. Seamless switching was thereby achieved, preventing overlapping or conflicting audio streams.

Additionally, a real-time data transmission and control system was established using Socket.IO, effectively connecting the robot voice module directly with a central control server. Robot behavior is dynamically adjusted based on signals of vision and payment system received from this server, toggling between passive advertisement and proactive customer engagement modes. To ensure greater reliability, reconnection logic was introduced to maintain continuous operation even in cases of network breakdowns or disconnections.

With these integrated technologies and diligent control of audio and network operations, the voice interaction module now delivers stable, easy-to-use conversational functionality to the robot, greatly enhancing overall user interaction and interaction quality.

G. Control Node and Integration

The robotics system is structured based on a central Master Node, which is implemented using Flask and Flask-SocketIO, to allow coordination between different hardware and software components. The architecture ensures seamless real-time communication across the system, managing state updates and coordinating interactions between the Computer Vision Module, NFC Payment Chip, Website UI, Voice Module, ROS-based Robot System, and the Servo-controlled Vending Mechanism. The Master Node holds a common state, keeping record of important variables like user interaction, navigation goal positions, detection confidence levels, etc.. Bi-directional communication with the Master Node and higher-level modules like the UI, voice system, and ROS framework is achieved using WebSockets, while direct hardware modules, like the NFC Payment Chip and Servo motors, are accessed via GPIO and I2C interfaces. Figure 6 shows the overall workflows of all nodes.

The Computer Vision Module runs in a separate multi-threaded environment with OpenCV to facilitate the recognition of human presence and relay detection outcomes, such as coordinates and confidence values, to the Master Node through a shared memory variable in raspberry pi. The data is then distributed to both the Voice Module and the ROS system to provide interactive response and autonomous navigation of the robot through sockets. In particular, ROS integration is accomplished via rosbridge, which is a WebSocket-based communications interface in which real-time movement commands as well as vision detected coordinates messages are passed back and forth in real-time utilizing pre-established topics such as `/cmd_vel` and `/goal_pos`. The Master Node constantly

reads detected positions and sends those messages to ROS so that the robot dynamically reacts to current environment.

The Voice Module interacts with the system through the processing of verbal input regarding transaction requests and confirmations. Upon detecting a potential customer, the master node will send a signal to voice module and conversation loop will start. Several speech commands and options can be called by the users, including voice selection of items and payment initiation. When no customer is detected, a signal is then transmitted to the voice module to start shouting and advertising, attracting more customers to the robot.

Another important part of the system is the payment mechanism based on NFC, which provides smooth and contactless transactions. NFC Payment Chip is interfaced with a Raspberry Pi directly using the I2C protocol, where payment confirmation is done using a polling mechanism that detects successful payments. After confirmation of payment received, the Master Node provides a vending signal to the Servo module using GPIO control, which triggers the dispensing mechanism. The circuit connection schematic is illustrated in Figure 5.

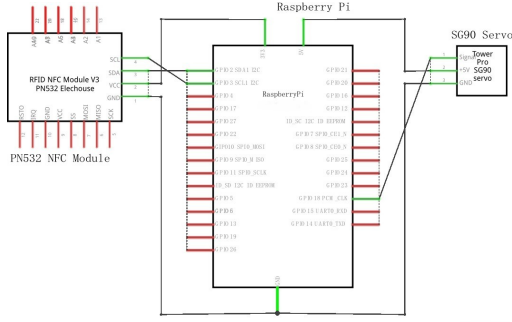


Fig. 5. Circuit Connection Schematics

The website-based UI serves as the primary interface for transaction management, allowing users to select products, confirm quantities, and complete payments. The UI maintains real-time synchronization with the Master Node through Web-Sockets, ensuring that all interactions are instantly reflected across the system. The payment confirmation process is directly linked to the NFC module, enabling vending operations upon successful transactions.

The overall integration (Figure 6) demonstrates a well-coordinated, real-time robotic system that seamlessly merges multiple subsystems to create an efficient, autonomous vending experience.

V. METHODOLOGY

A. Experimental Setup (Computer Vision)

We've separated the experiments into the following different scenarios:

- 1) Single person location detection. This tests the computer vision system's ability to detect the spatial position of a single person.
- 2) Multiple person detection. This test captures the locations of more than 1 person.

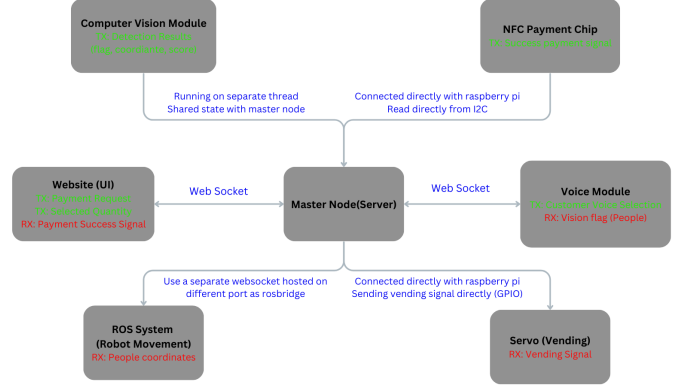


Fig. 6. Integration

- 3) Full detection. This test captures the locations and priority scores of multiple people. Including their face landmarks, hand gestures and poses.

B. Experiment and Data Collection (Integrated Parts)

The experimental grounds will be the EEE Building ground floor foyer and Dalby Court. The population sample will be university students. The camera and the UI system will detect how many people have interacted with the robot, and the payment system will record the number of successful sales. We measure the sales success rate by dividing successful sales by the total number of interactions. We do not actually charge the customers, only tricking them to think this is an actual vendor, keeping the test reliability.

- Low interactivity group: The robot is allowed to wander around and stop people, but it does not communicate with the customer. The customer is only able to complete the payment using the UI.
- Mid interactivity group: In addition to wandering around, robot interacts with the customer by responding with our preset fixed language base. It will be able handle basic verbal transactions. It does not alter its wordings regardless of different customer behavior.
- High interactivity group: The robot actively approaches customer and changes its interactions with the customer depending on the customer behavior.

For all the groups, the same items will be sold, the price will be the same. They will be at the test grounds for roughly the same time period.

The detailed results are documented in the next section.

VI. RESULTS AND DISCUSSION

A. Computer Vision

Experiment 1: Single-person full detection

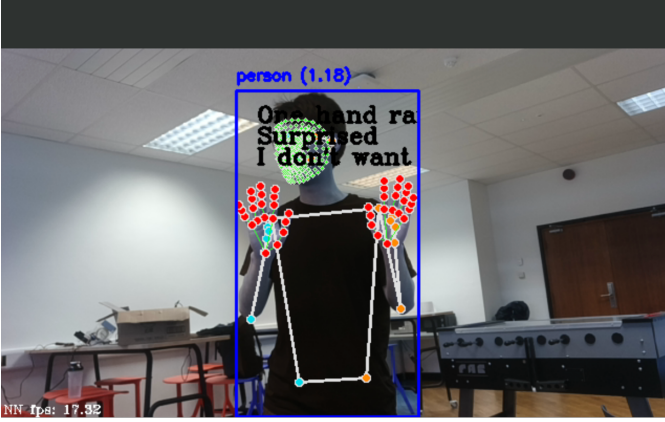


Fig. 7. Computer vision experiment. Capturing the bounding boxes, the facial landmarks, hand landmarks and the poses of a single customer.

Experiment 2: Multi-person full detection



Fig. 8. Computer vision experiment. Capturing the bounding boxes, the facial landmarks, hand landmarks and the poses of multiple customers.

Experiment 3: Spatial detection with ID table

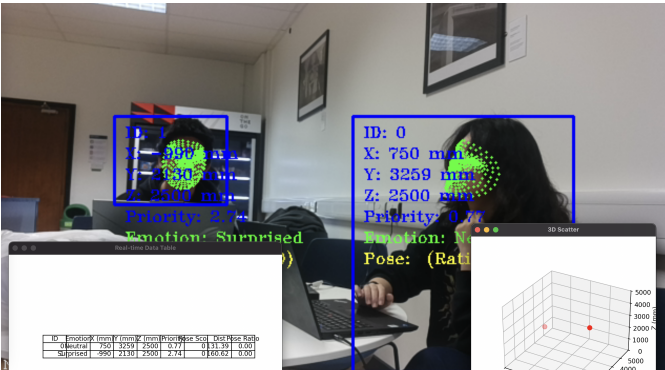


Fig. 9. Computer vision experiment. Capturing customer's location and display them in a real-time 3D graph and in a ID table

B. Integrated Experiment

1) *Experiment Overview:* The evaluation focused on comparing customer engagement and purchasing success between

two interaction levels: low interactivity and mid interactivity. Each group consisted of 10 participants, and their behavior upon noticing the robot was observed. The primary objective was to assess the success rate and how different interaction strategies influenced user engagement and purchasing behavior.

2) *Low Interactivity Group:* In the low interactivity setting, the robot did not engage in verbal communication. Most passersby were uncertain about its purpose, merely glancing at the robot before continuing on their way. Only 3/10 participants managed to figure out the purchasing process via the touchscreen interface, successfully completing a transaction. The lack of explicit guidance or verbal prompts resulted in minimal engagement, leading to a low conversion rate.

3) *Mid Interactivity Group:* For the mid interactivity group, the robot responded to users using a preset fixed language model, allowing for basic verbal transactions. This led to a noticeable increase in engagement—nine participants stopped to interact with the robot. However, the limitations of the fixed-response system reduced the overall effectiveness:

The robot failed to respond to two participants, causing them to disengage. This is due to the participants not knowing the correct key words the language model is looking for. Two additional participants attempted to make a purchase, but the robot misinterpreted their instructions and could not complete the transaction. Out of the nine customers who attempted a purchase, only five successfully received their item.

4) *Discussion and Insights:* The results indicate that increasing interactivity significantly improves customer engagement and purchase likelihood. While the low interactivity setting resulted in low engagement and confusion, the mid interactivity group saw a higher number of interactions, demonstrating that even basic verbal engagement makes a difference. However, the inability to handle diverse customer inputs reduced its effectiveness. We believe that a high interactivity version of the robot—incorporating an LLM-based response system tailored to the customer's language along with real-time sentiment analysis from computer vision—would significantly enhance engagement and purchasing success. This would enable the robot to adapt dynamically to user behavior, provide more natural interactions, and ultimately create a more seamless and intuitive vending experience.

VII. CONCLUSION AND FUTURE WORK

In this report, we presented the design and development of Movendor, an interactive, mobile vending robot aimed at revolutionizing traditional vending experiences. Unlike stationary vending machines, Movendor integrates advanced mobility, human-centered interaction, and intelligent perception systems to actively engage with potential customers. Movendor offers a dynamic and efficient vending solution, paving the way for the future of autonomous retail technologies.

A. Future Work

While Movendor demonstrates a significant step forward in autonomous vending, several areas can be further explored to enhance its functionality and user experience:

- **Integration of a Real Payment System:** Implementing secure and seamless payment solutions, such as NFC, QR code payments, and credit card processing, to enable real-time transactions.
- **Improved NLP for Voice Interaction:** Enhancing the natural language processing (NLP) system by incorporating a more powerful offline model, reducing latency, and enabling more fluid, responsive voice interactions with customers.
- **Expanded Inventory Management:** Currently Movender's inventory is limited to a few items with small size. It could be expanded to more inventories so the users have more choice.
- **Personalized Marketing Strategies:** Leveraging machine learning to analyze consumer preferences and suggest products dynamically based on past interactions.
- **Dynamic Gesture and Continuous Action Analysis:** Developing advanced computer vision models to recognize and interpret complex hand gestures and continuous movements, enabling more intuitive and seamless interactions.
- **Advanced Eye Tracking for Micro-Expression and Emotion Intensity Detection:** Implementing high-precision eye-tracking technology to analyze subtle micro-expressions, allowing for deeper insights into user emotions and engagement levels.
- **Fusion of Additional Sensor Data for Improved Navigation:** Integrating more sensor modalities, such as top-down sonar, to enhance autonomous navigation capabilities, improving obstacle avoidance and situational awareness.

By incorporating these advancements, Movender can evolve into a more intelligent and adaptable vending solution, bridging the gap between automation and human-like service in the retail industry.

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